

Pregnancy Risk Models

Validation on Maternal_Health_Risk_Data_Filled.csv

Data alignment

Maternal glucose converted g/L -> mg/dL (*100)
BodyTemp ignored
RiskLevel mapped to LOW/MEDIUM/HIGH

Rows

Maternal rows used: 1014
Synthetic split: 70/30 stratified
Four models compared

Compared Models

- Random Forest
- Logistic Regression
- SVM
- XGBoost

Diagnostics

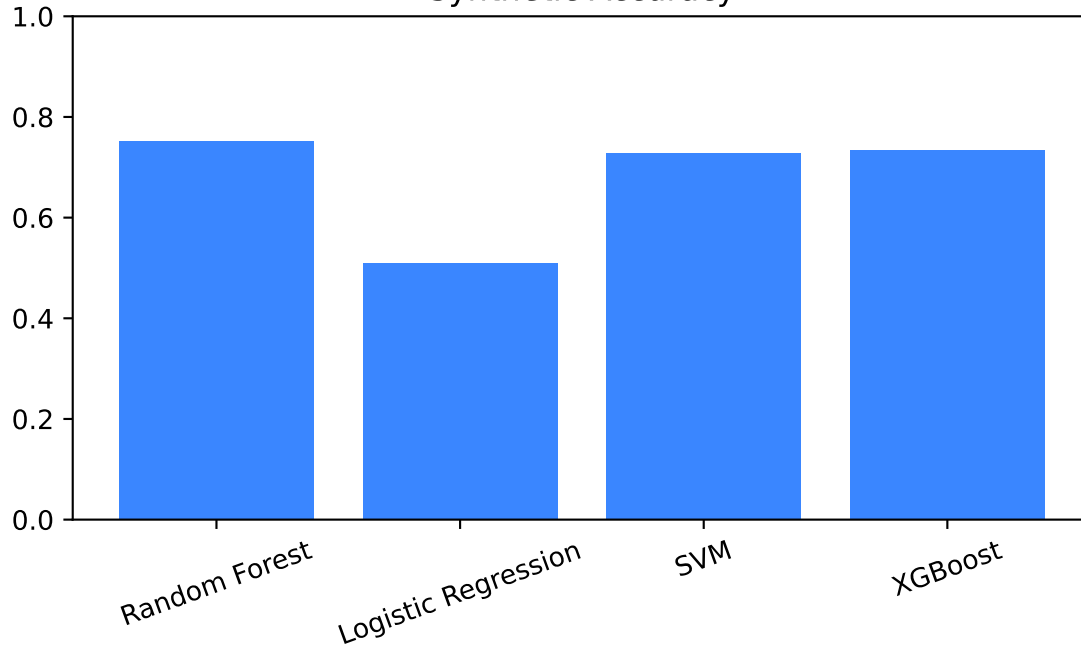
- Accuracy and Macro F1
- F1 per class
- Training time
- Strengths/weak points

Model Comparison Table

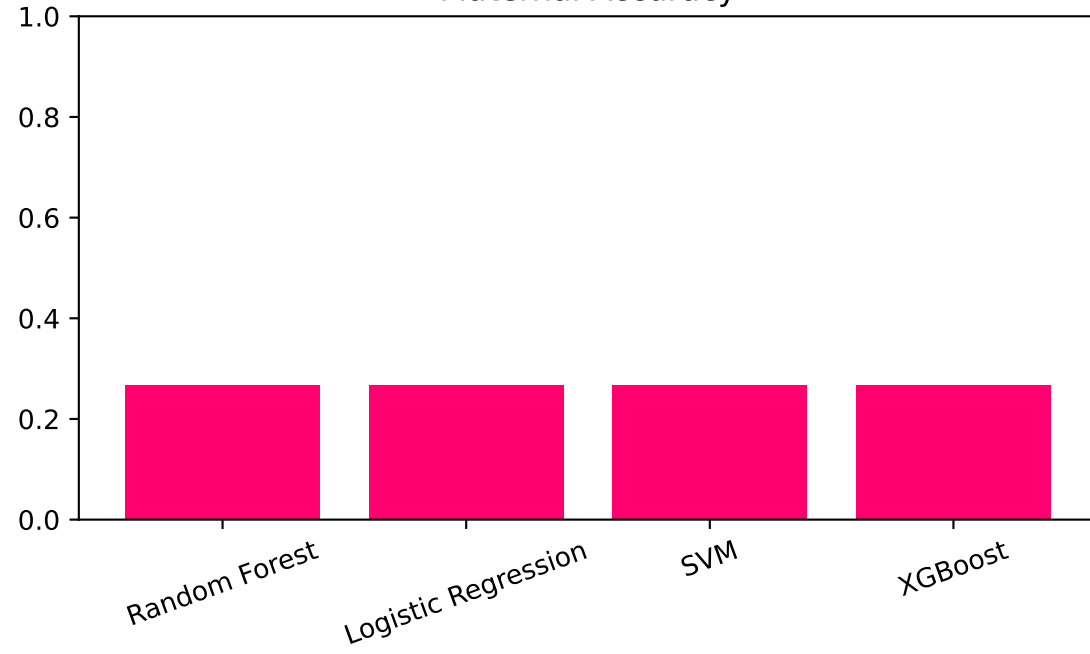
Model	Train(s)	Syn Acc	Real Acc	Real Macro F1	F1 HIGH	F1 LOW	F1 MEDIUM
Random Forest	1.303	0.752	0.268	0.141	0.423	0.000	0.000
Logistic Regression	0.036	0.510	0.268	0.141	0.423	0.000	0.000
SVM	1.033	0.728	0.268	0.141	0.423	0.000	0.000
XGBoost	0.692	0.733	0.268	0.141	0.423	0.000	0.000

Performance Dashboard

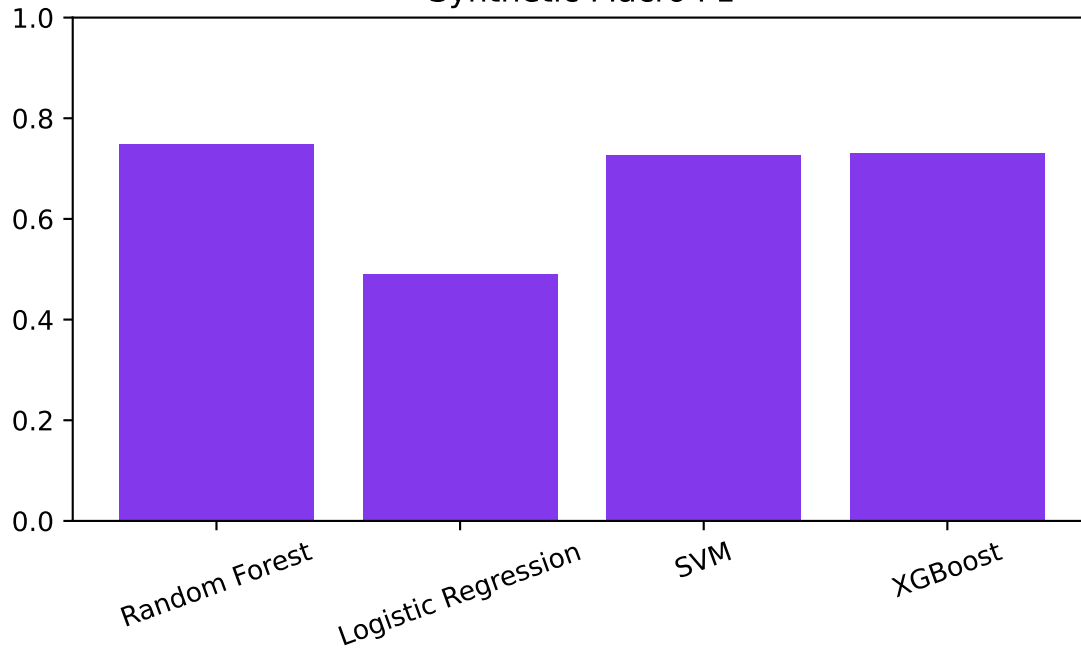
Synthetic Accuracy



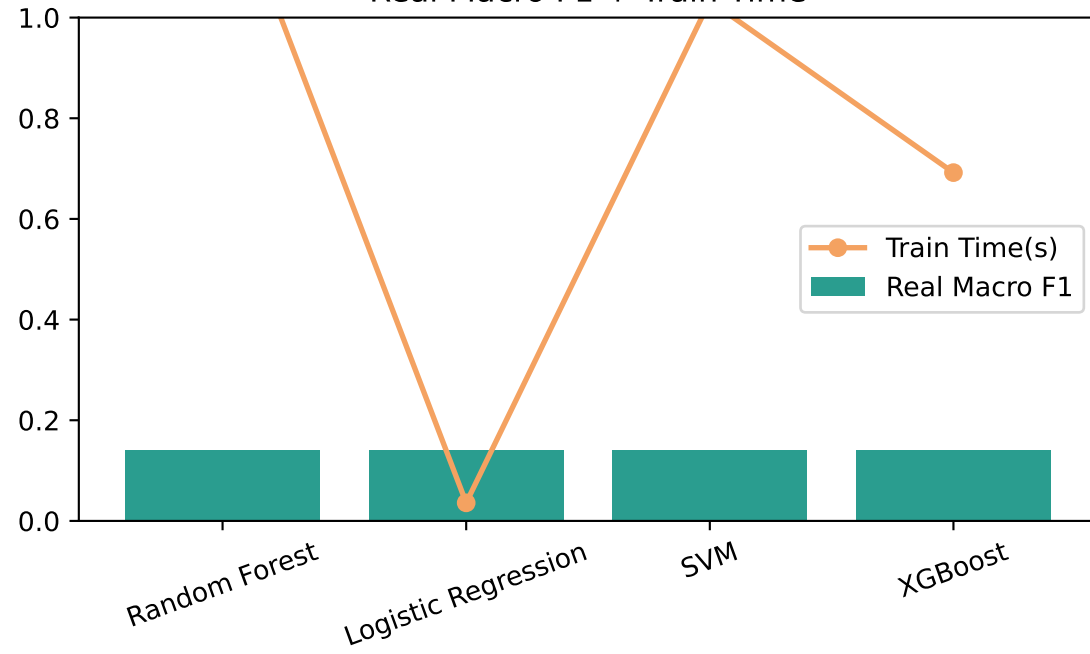
Maternal Accuracy



Synthetic Macro F1



Real Macro F1 + Train Time



Strengths and Weaknesses

Random Forest

Power points: Best F1 on HIGH: 0.423 | Best recall on HIGH: 1.000 | Best precision on HIGH: 0.268

Weak points: Weakest F1 on LOW: 0.000 | Weakest recall on LOW: 0.000 | Weakest precision on LOW: 0.000

Logistic Regression

Power points: Best F1 on HIGH: 0.423 | Best recall on HIGH: 1.000 | Best precision on HIGH: 0.268

Weak points: Weakest F1 on LOW: 0.000 | Weakest recall on LOW: 0.000 | Weakest precision on LOW: 0.000

SVM

Power points: Best F1 on HIGH: 0.423 | Best recall on HIGH: 1.000 | Best precision on HIGH: 0.268

Weak points: Weakest F1 on LOW: 0.000 | Weakest recall on LOW: 0.000 | Weakest precision on LOW: 0.000

XGBoost

Power points: Best F1 on HIGH: 0.423 | Best recall on HIGH: 1.000 | Best precision on HIGH: 0.268

Weak points: Weakest F1 on LOW: 0.000 | Weakest recall on LOW: 0.000 | Weakest precision on LOW: 0.000

Good and Not-Good Test Points

Random Forest

Good points: idx 123 true=HIGH pred=HIGH conf=0.99 ; idx 130 true=HIGH pred=HIGH conf=0.99

Not good points: idx 413 true=MEDIUM pred=HIGH conf=0.96 ; idx 787 true=MEDIUM pred=HIGH conf=0.96

Logistic Regression

Good points: idx 9 true=HIGH pred=HIGH conf=1.00 ; idx 20 true=HIGH pred=HIGH conf=1.00

Not good points: idx 74 true=MEDIUM pred=HIGH conf=1.00 ; idx 91 true=MEDIUM pred=HIGH conf=1.00

SVM

Good points: idx 0 true=HIGH pred=HIGH conf=0.00 ; idx 1 true=HIGH pred=HIGH conf=0.00

Not good points: idx 4 true=LOW pred=HIGH conf=0.00 ; idx 6 true=MEDIUM pred=HIGH conf=0.00

XGBoost

Good points: idx 123 true=HIGH pred=HIGH conf=1.00 ; idx 130 true=HIGH pred=HIGH conf=1.00

Not good points: idx 19 true=LOW pred=HIGH conf=1.00 ; idx 661 true=MEDIUM pred=HIGH conf=1.00